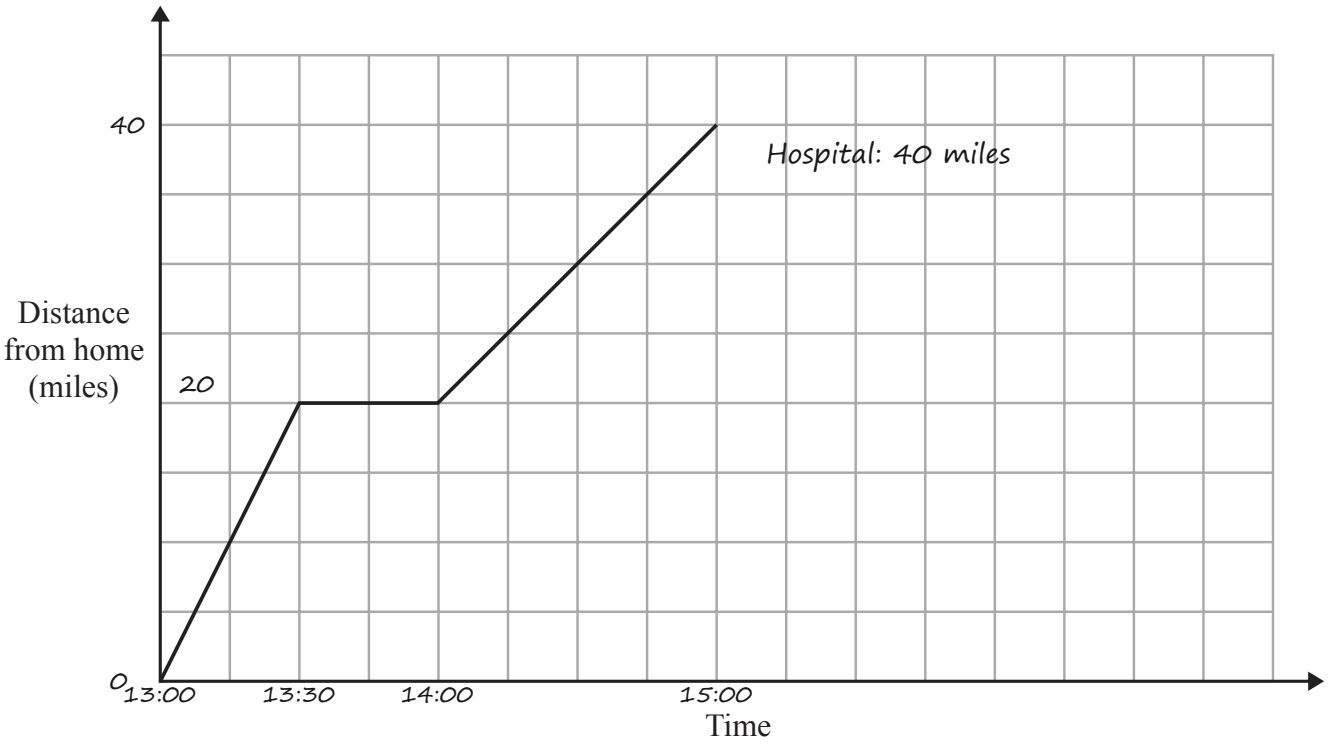
6 On Monday Ria delivered a parcel to a hospital.
The travel graph represents Ria's journey to the hospital.



Ria left home at 13 00
She drove for 30 minutes at a constant speed of 40 mph.
She then stopped for a break.

Ria then drove to the hospital at a constant speed.
She was at the hospital for 30 minutes.
She then drove home at a constant speed of 32 mph.

Show that she does not arrive home before 16 30

First leg: 40 × ½ = 20 miles.

1 mark
First leg = 20 miles

Hospital is twice that height: 40 miles from home.

1 mark
Scale original graph

Leave hospital at 15:00 + 30 min = 15:30.

Return time = 40 ÷ 32 = 1¼ hours = 1 h 15 min.

1 mark
1¼ hours

Home at 16:45, which is after 16:30.

1 mark
16:45; result shown

(Total for Question 6 is 4 marks)